

LLMs vs. Traditional Models in Complex Classification

A Comparison on Student Outcome Prediction

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Context: LLMs have revolutionized NLP, but their performance on structured tabular classification tasks is underexplored. Traditional ML models (e.g., Random Forest, Gradient Boosting), on the other hand, have been the backbone of classification tasks in various domains, including education.

⇒ Hypothesis: LLMs are not designed for classification tasks, especially with structured data, and may underperform compared to traditional ML models.

Problem Overview

Question: Can LLMs effectively perform complex classification tasks on structured data as well as traditional ML models?

Problem: Benchmarking LLMs (Qwen-2.5-3B, LLaMA-3-8B) against seven traditional ML models on a large, multiclass dataset.

- **Task:** Predict student outcomes (Distinction, Pass, Fail, Withdrawn) based on demographics, performance, and behavior features from the dataset.
- **Input:** 72 structured features (e.g., study time, assessment scores, VLE interactions).
- **Output:** Predicted class label for each student record.

Evaluation: Macro F1-score and Balanced Accuracy on a held-out test set.

Key Finding: LLMs heavily underperform traditional models on this structured classification task.

Background: Traditional ML Models (I)

Random Forest (RF)

- Ensemble of decision trees; each grown via random vector Θ
- Final prediction by majority vote
- Robust to overfitting; handles high-dimensional data

Support Vector Machine (SVM)

- Finds optimal separating hyperplane maximizing margin
- Kernel trick (RBF) for non-linear boundaries
- Effective in high-dimensional feature spaces

Gradient Boosting (GB)

- Sequential ensemble of weak learners (trees)
- Each tree corrects residuals of predecessors
- Final prediction: $F_M(x_i) = \sum_{m=1}^M \nu T_m(x_i)$

K-Nearest Neighbors (KNN)

- Non-parametric; classifies by majority of k nearest neighbors
- Distance metric: Euclidean (default)
- Error converges to $\leq 2 \times$ Bayes error rate as $n \rightarrow \infty$

Extreme Learning Machine (ELM)

- Single-hidden-layer feedforward network
- Random input weights; output weights solved analytically via Moore–Penrose inverse
- Fast training; competitive with SVM

Multilayer Perceptron (MLP)

- ≥ 3 layers (input, hidden, output)
- Learns via backpropagation & gradient descent
- Activation: sigmoid or ReLU

Large Language Models

- Transformer architecture (Vaswani et al., 2017); self-attention mechanism
- Pre-trained on massive corpora, then fine-tuned
- Excel at NLP; generalize across tasks without task-specific training

Models chosen:

- **Qwen-2.5-3B** (Alibaba) – optimized for structured I/O, coding, math
- **LLaMA-3-8B** (Meta) – optimized Transformer backbone, RLHF-aligned

Open University Learning Analytics Dataset – 32,593 student records, 20 raw features.

Module	Key Contents	Records
Student Demographics	Gender, region, education, IMD, age	32,593
Course Registrations	Module/presentation codes, length	22
Assessment Results	TMA/CMA/Exam scores, weights	173,912
VLE Interactions	Clickstream data (sum_click per site)	10,655,280

Target variable: `final_result` $\in$ {Distinction, Pass, Fail, Withdrawn}

Split: 80/20 train/test; SMOTE (ADASYN) applied to training set $\rightarrow$ 39,556 train records.

Three feature groups merged on `student_id`:

1. Background

- One-hot: gender, region, education, disability
- Ordinal: age band, IMD band

2. Performance

- Merge assessments on `id_assessment`
- Impute missing scores (median)
- Engineer: submission delay, late/early flags, weighted score
- Aggregate per student

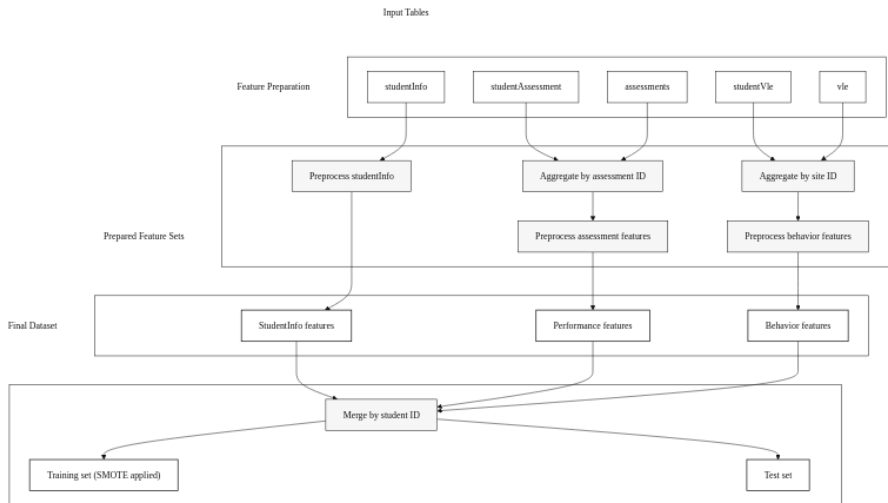
3. Behavior

- Cutoff: latest assessment date (prevent leakage)
- Merge VLE clicks with activity type
- Pivot into wide features (e.g., `clicks_forum`)
- Chunk-level aggregation

Final dataset: 32,593 records $\times$ 72 features (including ID and label).

Preprocessing Pipeline (cont.)

Visualization:



Seven models evaluated: RF, SVM, GB, DT (CART + C5.0), KNN, ELM, MLP.

Selection rationale:

- RF & SVM: top classifiers per Delgado et al.'s systematic review
- DT, GB, ELM, MLP: strong performers in educational data mining literature
- KNN: baseline to test if simple distance-based methods suffice

Training procedure:

- 1 5-fold cross-validation on training set for hyperparameter tuning
- 2 Select best hyperparameters by average F1-score across folds
- 3 Train final model on full training set with selected hyperparameters

Implementation: `scikit-learn` (Python); C5.0 via R.

Methodology: LLMs & Prompting

Models: Qwen-2.5-3B and LLaMA-3-8B (from Hugging Face), **no fine-tuning**.

Prompting techniques:

- **Zero-shot:** Classify directly from feature description alone
- **One-shot:** One labeled example provided before the query
- **Five-shot:** Five labeled examples provided before the query

Prompt structure:

- 1 Base instruction + allowed labels (Distinction, Pass, Withdrawn, Fail)
- 2 Few-shot examples (if any): student record JSON → LABEL=<class>
- 3 Target student record
- 4 Strict output format: LABEL=<one of four classes>

Key constraint: LLMs must infer patterns from examples only – no gradient-based learning on training data.

Results: Traditional ML Models

Metrics: Accuracy, Balanced Accuracy, Macro F1.

Model	Test Accuracy	Balanced Accuracy	Macro F1
Decision Tree	0.7188	0.6850	0.6752
Random Forest	0.7612	0.7193	0.7199
Gradient Boosting	0.7647	0.7178	0.7220
Support Vector Machine	0.7400	0.6354	0.6524
K-Nearest Neighbors	0.6872	0.5954	0.6006
MLP Classifier	0.7274	0.6760	0.6785
ELM Classifier	0.7289	0.6300	0.6448

Top performers: Gradient Boosting leads in test accuracy and Macro F1, while Random Forest is best on balanced accuracy.

CV results confirm RF and GB have highest mean balanced accuracy (~ 0.82 – 0.84) with low variance.

Results: Large Language Models

Metric: Macro F1 and Balanced Accuracy.

Qwen-2.5-3B

- 0-shot: Macro F1 = 0.2178, Bal. Acc. = 0.2970
- 1-shot: Macro F1 = 0.3867, Bal. Acc. = 0.5621
- 5-shot: Macro F1 = 0.3056, Bal. Acc. = 0.3773

LLaMA-3-8B

- 0-shot: Macro F1 = 0.2802, Bal. Acc. = 0.3414
- 1-shot: Macro F1 = 0.3919, Bal. Acc. = 0.5086
- 5-shot: Macro F1 = 0.4942, Bal. Acc. = 0.4811

Key observations:

- All LLM results remain below traditional ML on balanced accuracy and Macro F1
- 1-shot gives the best balanced accuracy for both models (Qwen: 0.5621, LLaMA: 0.5086)
- LLaMA benefits from 5-shot in Macro F1 (0.4942), while Qwen peaks at 1-shot

Interpretation

- Traditional models (esp. GB) effectively exploit structured features – study materials access and study time are strong predictors
- LLMs suffer from **fallthrough to default heuristics**: they collapse predictions to one or two dominant classes
- General-purpose LLMs are not suited for structured tabular classification without fine-tuning
- Findings support traditional ML as the reliable choice for early-warning systems in education

Threats to Validity

- OULAD lacks external factors: student mental health, personal emergencies (“black swans”)
- Risk of overfitting to outdated academic contexts; bias toward past education systems
- **Hawthorne effect**: students aware of monitoring may alter behavior, distorting predictions
- LLM evaluation limited to two small-parameter models with basic prompting

Conclusion:

- Traditional ML models (RF, GB) significantly outperform LLMs on structured student outcome classification
- LLMs (Qwen-2.5-3B, LLaMA-3-8B) achieve $<50\%$ balanced accuracy vs. $\sim 80\%$ for GB
- LLMs' NLP strength does not transfer to tabular classification tasks
- Result reinforces traditional ML as the primary vessel for classification in educational data mining

Future Work:

- Refine prompting strategies (chain-of-thought, retrieval-augmented)
- Test larger, more capable LLMs (70B+ parameter models)
- Incorporate richer socio-economic data and cross-discipline datasets
- Explore fine-tuning LLMs on domain-specific tabular data